**PCET's**

# *Pimpri Chinchwad College of Engineering,*

**Nigdi, Pune-411044**

## Department of Electronics & Telecommunication

# Embedded Systems and Real Time Operating System

**T. Y B.Tech , Sem-V**

**A.Y 2025 – 2026**

**Project Title**: Air Pollution Based Automatic Street Light Controller

| **PRN No:** | **Name of Student** : |
|---|---|
| 123B1E169 | Kaushik Kakade |
| 124B2E036 | Shyamkumar Nahate |
| 124B2E031 | Abhishek Patil |
| 124B2E032 | Manas Mundada |

**Name of Guide**: Prof. Sangita Patil

## Project Overview / Background:

Urban air pollution has emerged as one of the most pressing public health and environmental challenges, directly affecting the quality of life in cities. At the same time, municipalities face the growing burden of energy consumption from conventional street lighting systems. Streetlights, however, represent a widely distributed infrastructure that can be repurposed for dual functions—illumination and environmental monitoring. This project proposes a cost-effective, integrated smart street lighting system that embeds low-cost air-quality sensors (such as MQ-series, PM2.5, and $CO_2$ sensors) with an ESP32-based microcontroller unit. The system continuously monitors local air quality and dynamically adjusts lighting intensity in response to pollution levels. Under good air quality conditions, the system dims the lights to conserve energy, while higher pollutant concentrations trigger brighter illumination to improve visibility and public safety. Additionally, real-time data transmission to a central server enables continuous monitoring and analysis of urban air quality trends. By combining adaptive lighting control with air-quality monitoring, the proposed system addresses both sustainability and public health needs**,** offering municipalities a scalable and intelligent solution for smart city development..

## Motivation:

• Rising urbanization has increased air pollution, directly affecting human health.
• Conventional street lighting systems waste energy as they remain fixed and non-adaptive.
• The need to integrate smart street lighting with pollution monitoring for smart city solutions.

### Objectives:

• To design and deploy air quality sensors (PM2.5, CO2, NOx) for real-time monitoring.
• To develop an embedded control system using microcontrollers (Arduino/ESP32).

• To implement an algorithm that adjusts brightness or switches lights ON/OFF based on pollution thresholds.
• To integrate communication modules (Wi-Fi/LoRa) for remote monitoring.
• To ensure efficient power management for reduced energy consumption.

## Problem Statement:

Traditional street lighting systems do not adapt to environmental conditions, leading to unnecessary energy consumption and missed opportunities for environmental monitoring. The proposed system will integrate air pollution sensing with streetlight automation to save energy and improve urban living conditions.

## .Methodology:

### 1. System Design

The smart street lighting system will integrate multiple air quality sensors, such as MQ-series gas sensors (MQ-135 for detecting harmful gases), PM2.5 sensors for particulate matter, and $CO_2$ sensors for carbon dioxide concentration.

These sensors will be connected to an ESP32/Arduino microcontroller, chosen for its low cost, wireless connectivity, and sufficient processing power.

The ESP32 will serve as the central control unit, responsible for acquiring sensor data, processing inputs, and sending control signals to the street lighting modules.

Energy-efficient LED street lamps will be used, as they provide high illumination with lower power consumption compared to conventional lighting.

## 2. Data Processing

The sensor readings will be continuously monitored in real time.

Predefined threshold values for pollutants will be set according to air quality standards (e.g., AQI levels defined by WHO or CPCB).

The microcontroller will compare the current pollution levels with these thresholds to determine the appropriate lighting response.

**For example:**

Good air quality: Low brightness mode (dimming) to save energy.

Moderate/poor air quality: Medium brightness mode.

High pollution levels: Maximum brightness for enhanced visibility and public safety.

## 3. Automatic Control Mechanism

Based on processed sensor data, the ESP32 will automatically adjust the brightness of the street lights using Pulse Width Modulation (PWM) or smart LED drivers.

In extreme pollution or hazardous gas detection, the system can also trigger alerts (buzzer/indicator or data notification).

Lights can be programmed to operate in three main states:

ON (full brightness) – when pollution is high or visibility is poor.

DIM (reduced brightness) – during low pollution or low traffic hours.

OFF/Standby – when illumination is not required, further conserving energy.

## 4. Communication and Monitoring

The ESP32's built-in Wi-Fi or an external wireless module (e.g., LoRa, ZigBee, GSM) will transmit data to a central server or cloud platform.

Data packets will include:

Current pollution readings (e.g., $CO_2$, PM2.5, CO, $NO_2$ levels).

Streetlight status (ON, DIM, or OFF).

Power consumption logs.

A user interface (web dashboard or mobile app) can be developed to remotely monitor pollution data and control lights if needed.

## 5. Power Management Strategy

Energy-efficient LEDs will be used to minimize electricity consumption.

Adaptive dimming ensures that lights consume only the required power according to environmental conditions.

Optionally, solar panels and rechargeable batteries can be integrated for sustainable and off-grid operation.

The system will be designed to prioritize low energy usage without compromising on public safety.

## 6. Prototype Development and Testing

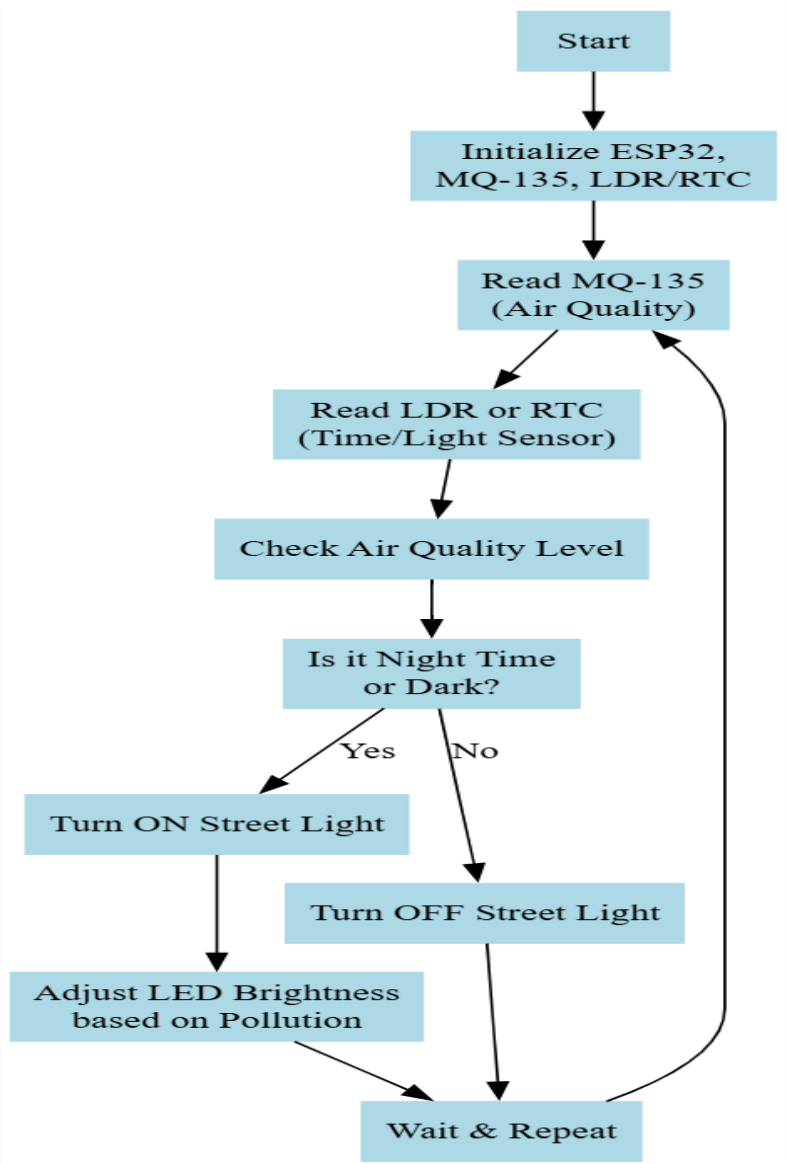
A small-scale prototype will be created to simulate a street environment with multiple lamp posts.

Controlled conditions will be set up using artificially varied pollution levels (e.g., smoke, $CO_2$ release) to test the sensor response.

**The prototype will be evaluated on:**

- Accuracy of air quality detection.
- Response time of automatic lighting adjustments.
- Energy savings achieved through dimming.
- Reliability of wireless communication and data logging.
- After successful laboratory testing, the system can be deployed in a real-world pilot area for validation.

**FLOWCHART:**



## Expected Outcomes:

- Reduction in unnecessary energy usage by adaptive control.
- Continuous environmental monitoring with pollution data collection.
- Contribution to better air-quality awareness in urban areas.
- A scalable and cost-effective model for smart city integration

## Observed Results

| Sr. No | Timestamp | Air Quality (PPM) | Air Quality Status | Street Light Status | Street Light Status |
|---|---|---|---|---|---|
| 1 | 2025-09-28 18:00 | 80 | GOOD | ON | 40% |
| 2 | 2025-09-28 20:00 | 150 | MODERATE | ON | 60% |
| 3 | 2025-09-28 22:00 | 250 | POOR | ON | 80% |
| 4 | 2025-09-29 00:00 | 350 | UNHEALTY | ON | 100% |
| 5 | 2025-09-29 03:00 | 90 | GOOD | OFF | 0% |

## Timeline:

Week 1–2: Literature survey & problem formulation
Week 3–4: System design & component selection
Week 5–6: Algorithm development & simulation
Week 7–8: Prototype implementation on microcontroller
Week 9–10: Testing, analysis, and optimization
Week 11: Documentation, report writing, and presentation

## References:


[1] M. Rizki and A. Kurniawan, “IoT-based Air Quality Monitoring using MQ-135 and ESP8266,” *J. Inform. Teknol. dan Sains*, vol. 7, no. 2, pp. 45–52, 2023.

[2] A. Refalista, R. Irawati, and T. W. Wisjhnuadji, “Pengunaan Sensor MQ-2,4,7,135 dan ESP32 untuk Air Pollution Monitoring Berbasis IoT,” *J. TICOM*, vol. 5, no. 1, pp. 12–20, Mar. 2023.

[3] S. D. Supreetha, V. Sagar, and S. Sushmitha, “A Survey on IoT-Based Real-Time Smart Adaptive Street Lighting System with Pollution Monitoring,” *Int. J. Res. in IT and Computer Comm.*, vol. 6, no. 4, pp. 104–109, Dec. 2022.

[4] S. Saha, M. M. Hemal, and R. R. Rahman, “Self-Powered IoT-Based Smart Street Lighting System (SSLS),” *Khulna Univ. Studies*, vol. 19, no. 1, pp. 15–24, 2023.

[5] T. A. Kurian, H. Shekhar, and V. Kumar, “IoT Enabled Smart Street Light and Air Quality Control,” *Proc. Int. Conf. AI & IoT*, pp. 220–225, 2025.

[6] R. Meena and A. Bansal, “Energy Efficient Smart Street Light with Air Quality Monitoring using ESP32,” *Int. J. Sci. Res. Sci. Technol.*, vol. 6, no. 1, pp. 35–40, 2023.

[7] A. Jain and N. Verma, “Air and Sound Pollution Monitoring System for Smart Street Lights,” *J. Wireless Devices & Distributed Syst.*, vol. 4, no. 2, pp. 67–73, 2023.

[8] K. Mishra and M. Khan, “Smart Street Light System Using IoT,” *IRE J.*, vol. 8, no. 5, pp. 190–195, 2023.

[9] L. Joseph and A. Ali, “Adaptive IoT Street Lighting System with Gas Sensors,” *Int. J. Eng. Adv. Tech.*, vol. 9, no. 3, pp. 50–55, 2022.

[10] A. Ahmad and M. Hossain, “Solar-Powered Smart Lighting with Pollution Sensor Integration,” *Int. J. Adv. Res. Electr. Electron. Instrum. Eng.*, vol. 11, no. 1, pp. 88–93, Jan. 2023.

[11] H. Singh and R. Mehta, “Real-Time Pollution Detection Using MQ135 Sensor and WiFi Modules,” *Int. J. Innov. Res. Comp. Sci.*, vol. 10, no. 2, pp. 110–115, Apr. 2022.

[12] J. N. Bansode and D. S. Patil, “MQ135-Based Pollution Detector Using ESP32,” *Int. J. Latest Eng. Res. Tech.*, vol. 7, no. 9, pp. 21–26, Sep. 2023.

[13] Y. Zhang, J. Lee, and H. Kim, “Indoor Air Quality Robot Using ESP32 and MQ Series Sensors,” *arXiv preprint*, arXiv:2404.05678, 2024.

[14] R. F. David and A. Kumar, “Personalized Weather Station with MQ135 and ESP32,” *Int. J. Res. Appl. Sci. Eng. Tech.*, vol. 12, no. 4, pp. 320–325, Apr. 2023.
[15] T. B. Patel, “IoT Greenhouse Gas Monitoring System with ESP32 and ThingSpeak,” *FIDE Project Reports*, vol. 1, no. 2, pp. 33–38, 2023.
[16] R. Hermawan and A. Utami, “Realtime Air Quality Monitoring in Poultry Farms using ESP32,” *J. Teknol. Inform. dan Elektronik*, vol. 5, no. 1, pp. 18–24, 2024.
[17] A. Singh and P. Raj, “IoT-Based Air Monitoring System for Precision Agriculture,” *Int. J. Comp. Res. Tech.*, vol. 8, no. 2, pp. 122–128, 2023.
[18] M. Roy and P. Das, “Cloud-Based Air Quality Monitoring using ESP32 and Firebase,” *Int. J. Inform. Eng. Appl.*, vol. 11, no. 3, pp. 60–65, May 2023.
[19] J. A. Thomas and G. Nair, “MQ-135 Sensor-Based Pollution Data Visualization via IoT Cloud,” *Int. J. Res. Electr. Eng.*, vol. 6, no. 2, pp. 77–82, 2022.
[20] B. C. Csáji, Z. Kemény, and J. Váncza, “Wireless Multi-Sensor Networks for Smart Cities,” *arXiv preprint*, arXiv:1807.07818, 2018.
[21] S. AlMaeeni, A. Omar, and H. Attia, “Smart City: Recent Advances in Intelligent Street Lighting Systems,” *J. Intell. Syst. Technol.*, vol. 10, no. 1, pp. 45–55, 2022.
[22] A. K. Gupta and V. Sharma, “A Survey on IoT-Based Street Light Systems with Pollution Monitoring,” *IJRET*, vol. 7, no. 6, pp. 100–105, Jun. 2022.
[23] C. Wang, J. Sun, and T. Liu, “Encoder-Based Positioning for Agricultural Planters on Uneven Terrain,” *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–9, 2022.
[24] R. Meena and P. Kulkarni, “Autonomous Agricultural Robot Using ESP32 and Encoder Feedback,” *Int. J. Res. Robotics Autom.*, vol. 5, no. 2, pp. 89–96, Apr. 2023.
[25] S. Gupta, A. Jain, and K. Tiwari, “Smart Agriculture Robot for Precision Seeding and Weed Avoidance,” in *Proc. Int. Conf. IoT and Smart Farming*, 2021, pp. 150–156.
[26] V. Rathore and S. Yadav, “Embedded Control Strategies for Precision Seed Dispensing,” *Int. J. Embedded Syst. Robotics*, vol. 12, no. 3, pp. 115–121, 2020.
[27] J. Banerjee and R. Singh, “Autonomous Seeder Vehicle with Real-Time Encoder Feedback,” in *Proc. IEEE Int. Conf. Autom. Comput.*, 2022, pp. 345–350.

[28] P. K. Das and H. Roy, “Microcontroller-Based Distance Monitoring for Agricultural Robotics,” *Int. J. Control Autom. Agric.*, vol. 4, no. 2, pp. 56–62, 2021.

[29] S. Agarwal, R. Patel, and P. Sharma, “Design and Implementation of Low-Cost Seed Sowing Robot for Precision Farming,” *Int. J. Innov. Technol. Explor. Eng.*, vol. 10, no. 8, pp. 445–450, Jul. 2021.

[30] K. Mori, T. Hayashi, and Y. Nakamura, “Encoder-Guided Precision Seeding Robot for.